

THE DATA-DRIVEN DECISION TO ELECT A MAC OVER A WINDOWS LAPTOP

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1. INTRODUCTION

I have been using a Dell Latitude E7240 (64 bit, Intel Core i7-4600U @ 2.10 GHz, 16 MB RAM, 12 inch touch screen, 250 GB HD) running Windows 7 Enterprise SP1, borrowed from Department 5421 OAA Valerie Marquez. Instead of ordering a laptop computer immediately upon my first day of work (over a month ago), a delay, I opined, would allow me time to collect information and assess; to make a decision driven by data, not just inertia (“that’s how we have always done it”) or emotion (“that’s how I feel”).

2. ASSESSMENT

2.1. The Microsoft Office hurdle. During this assessment time, I collected data from many sources. My manager, Doug Dederman, expressed a preference for a PC running Windows over Mac because he had encountered Mac users who suffered from incomplete or absent functionality with the Microsoft Office suite of software, in particular, with the calendar and email programs. My colleague, Candice Cooper, stated that she had encountered incompatibilities with the PowerPoint application on a Mac. Some PowerPoint documents, created or last written to file from a PC, when opened on a Mac, had equations that were mangled, suggesting incompatibility between the Mac and PC implementation of Equation Editor.

I noted that these two experiences were historical, and wanted to determine if these bugs persisted, or had been fixed.

I tested functionality on and compatibility between a Mac and a PC running Microsoft Word 2016, Outlook 2016, and PowerPoint 2016 using a Microsoft Office 365 subscription. This testing failed to reproduce the aforementioned deficiencies in functionality and compatibility. The result of this point testing, while anecdotal, persuades me that hurdles of the past remain in the past, and are not applicable to the present-day software offerings.

In addition, data from ALEGRA developers Glen Hansen, Duane Labreche, and Jason Sanchez, all Mac users in the Sandia desktop computer ecosystem, indicated no loss of functionality and compatibility in their present-day experience with Microsoft products.

Finally, Microsoft’s website for this suite of products speaks to Office 365 compatibility not only between Mac and PC, but also across mobile devices (iOS, Android, Windows).

2.2. The runs-only-on-Windows hurdle. Applications that run only on the Windows operating system may present worry for those who run on a Mac instead of a PC. Indeed, I encountered this problem with at least three applications used at my engineering consultancy. To surmount these hurdles, I used VMware Fusion Pro for Mac, which allows users to virtualize the Windows operating system, and thus run Windows-only applications concurrently on a Mac. I used this hybrid system, the baseline Mac hardware running OS X (now, macOS Sierra) with a virtualized installation of Windows 7 (and later, version 10), for years, with complete satisfaction. A Mac running virtualized Windows was the ultimate “have your cake and eat it too” solution.

2.3. Focus on Analyst/Developer Use Cases. In the context of ALEGRA development, there are several compelling reasons to choose a Mac over a PC. First, ALEGRA can be built on a Mac; it

cannot be built on a PC. So, the use case of an analyst/developer engaging at a client site and, for example, updating a material model on-the-fly is lost on a PC, found on a Mac. Similarly, for any use case that is network-deprived (fly on an airplane, work at a remote site, work where the network connection is not secure or suffers frequent down time, to name four), a Mac can keep a local build of ALEGRA available to the user, a PC cannot.

Why does ALEGRA run on a Mac? Because the Mac operating system is a descendant of the Berkeley Software Distribution (BSD) UNIX. And UNIX is the second compelling reason to choose a Mac instead of a PC. The target operating system for ALEGRA is Linux, and the target compiler is `gcc`. Because Mac UNIX and Linux (specifically, Red Hat Enterprise Linux used on the HP workstation) descend from the same Portable Operating System Interface (POSIX) and AT&T UNIX System V Interface Definition (SVID) parents, integration of shells, scripts, and infrastructure between Mac and Linux are tightly integrated, if not essentially invisible. In contrast, Windows requires the user to bolt on and maintain compensatory applications, such as VIM for Windows, PuTTY for Window, and Cygwin for Windows, to bring Microsoft's operating system environment up to snuff.

There are many other user productivity reasons that make Mac the preferred desktop client for ALEGRA development. While not exhaustive, the following example is illustrative.

A text file generated on a PC will insert end-of-line (EOL) characters that can impose problems for the Mac or Linux user. Text files created on Windows will have a pair of carriage return (CR) and line feed (LF) characters to terminate lines. UNIX uses a LF only. Text files created on Windows will display on UNIX with the extra CR displayed as `^M` or `<cr>` at the end of the line or as a second line break.

A common workaround, when transferring files between client and server with file transfer protocol

(FTP), is to use an ascii transfer, which converts line breaks, instead of a binary transfer, which preserves the native line break characters.

While this and other workarounds exist to tame PC-to-Linux translation problems, avoiding such hoop jumping in the first place is a better practice. Indeed, this is the thematically consistent feedback I have received from numerous ALEGRA developers: If you have the choice, choose Mac. If you don't have the choice, you can still make the PC work, with additional effort above and beyond what is required with a Mac.

Sand in the gears, then, is the proverbial description of PCs in the ALEGRA high performance computing environment at Sandia. And if PC clients are sand, then Mac clients are surely viewed by ALEGRA developers as no less than the oil.

3. RECOMMENDATION

MacBook Pro with Retina display, 15 inch, 2.5 GHz Quad-core Intel Core i7, 16 GB 1600 MHz DDR3L SDRAM, 512 GB PCIe-based flash storage. $\square$